

1. Administrative & Study Identification

Full Study Title: A Study of Multiple Immunotherapy-Based Treatment Combinations in Hormone Receptor (HR)-Positive Human Epidermal Growth Factor Receptor 2 (HER2)-Negative Breast Cancer
Short Title/Acronym: MORPHEUS-HR+BC **Protocol Number:** B039606 **NCT Number:** NCT03280563 **Posting ID:** 347634327880628897 **Sponsor Name:** Roche / F. Hoffmann-La Roche Ltd **Investigational Product:** Atezolizumab (Tecentriq) in combination with targeted/endocrine agents (e.g., ipatasertib, entinostat, abemaciclib, samuraciclib, giredestrant, fulvestrant, tamoxifen) **Phase of Development:** PHASE1, PHASE2 (Phase Ib/II) **Trial Status:** COMPLETED **Medical Monitor:** Hoffmann-La Roche Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy (Objective Response Rate [ORR]) and safety/tolerability of several immunotherapy-based combination treatments in participants with HR-positive, HER2-negative breast cancer.
Secondary Objectives: To evaluate Progression-Free Survival (PFS), Overall Survival (OS), Duration of Response (DOR), Clinical Benefit Rate (CBR), and pharmacokinetics.

2.2 Background & Rationale To address the unmet medical need in metastatic or inoperable locally advanced HR-positive, HER2-negative breast cancer patients who have progressed during or following treatment with a cyclin-dependent kinase (CDK) 4/6 inhibitor. Combining cancer immunotherapy (atezolizumab) with targeted agents or endocrine therapy aims to overcome acquired resistance, target multiple cancer immune escape mechanisms simultaneously, and extend durable clinical benefit to more patients.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Umbrella Platform Trial) **Control Type:** Active Comparator (Fulvestrant monotherapy in Stage 1) **Blinding:** Open-label **Randomization:** Randomized to either a control arm or one of multiple experimental doublet/triplet combination arms.

3.2 Planned Enrollment & Duration Target **NS:** 144 **Number of Sites:** Global multicenter trial **Estimated Study Start:** 2017 **Estimated Primary Completion:** Completed

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Metastatic or inoperable locally advanced, histologically or cytologically confirmed invasive HR-positive, HER2-negative breast cancer. 2. Disease progression during or after first- or second-line hormonal therapy with a CDK4/6 inhibitor (e.g., palbociclib, ribociclib, abemaciclib). 3. Measurable disease per RECIST v1.1 and an Eastern Cooperative Oncology Group (ECOG) performance status of 0-1 (for Stage 1).

4.2 Exclusion Criteria 1. Prior fulvestrant or cytotoxic chemotherapy for metastatic breast cancer (for Stage 1). 2. Active autoimmune disease, positive human immunodeficiency virus (HIV), active Hepatitis B or C, or active tuberculosis. 3. History of malignancy other than breast cancer within 2 years prior to screening (except those with negligible risk of metastasis/death).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Atezolizumab is given at 840 mg via intravenous (IV) infusion on Days 1 and 15 of each 28-day cycle, or at 1200 mg IV on Day 1 of each 21-day cycle, depending on the specific targeted/endocrine partner agent combination. **Route of Administration:** Intravenous (atezolizumab) + Oral or Intramuscular components (partner targeted/endocrine agents). **Associated Drug Entities:** Atezolizumab, fulvestrant, and other related/background endocrine therapies such as Tamoxifen (ATC Code: L02BA01; Trade names: Soltamox, Noltam, Barr - Tamoxifen Citrate, Nolvadex, Emblon, Fentamox, Nolvadex-D, Nolvadex Forte, Oestrifin, Tamofen). **Duration of Treatment:** Continued until unacceptable toxicity, loss of clinical benefit as determined by the investigator, or disease progression.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care. **Prohibited:** Concurrent administration of other experimental therapies or antineoplastic agents outside of the study protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, RECIST baseline imaging, Labs, Biomarker/Genomic Tests
Baseline	Day 0	Randomization, First Dose Administration
Interim	Every 21 or 28 Days	Atezolizumab infusions, Safety Labs, ongoing Tumor assessments (every 8-12 weeks)
End of Study	At Progression / Loss of Benefit	Final assessments, Exit Interview, Safety Follow-up, Transition to Stage 2 (if eligible)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Objective Response Rate (ORR) and incidence/severity of adverse events. **Measurement Method:** Investigator-assessed tumor response according to Response Evaluation Criteria in Solid Tumors (RECIST) v1.1.

7.2 Secondary & Exploratory Endpoints Progression-Free Survival (PFS) and Overall Survival (OS). Duration of Response (DOR) and Clinical Benefit Rate (CBR).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard clinical monitoring, with specific attention to immune-related Adverse Events of Special Interest (AESI) for atezolizumab, such as pneumonitis, autoimmune myelitis, pericardial disorders, and facial paresis. **Serious Adverse Events (SAEs):** Standard 24-hour reporting. **Data Monitoring Committee (DMC):** Platform features dynamic safety/efficacy monitoring to add new combination arms or close existing ones displaying unacceptable toxicity or lack of efficacy.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy outcome analysis. Safety-evaluable population for all AE/SAE evaluations. **Sample Size Justification:** Target enrollment of 144 participants; relies on an adaptive, multiple-cohort umbrella trial design to identify early signals of activity. **Handling of Missing Data:** Missing tumor assessments are evaluated as non-evaluable (NE) per RECIST criteria.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Sponsor-directed onsite and remote monitoring per protocol. **Audit Trail:** Validated eCRF database with standard data clarification workflows.

11. Study Identifiers & Governance

Primary ID: {{B039606}} **Registry Number:** {{NCT03280563}} **Posting ID:** {{347634327880628897}} **Sponsor/Lead Organization:** {{Roche / F. Hoffmann-La Roche Ltd}} **Study Title:** {{A Study of Multiple Immunotherapy-Based Treatment Combinations in Hormone Receptor (HR)-Positive Human Epidermal Growth Factor Receptor 2 (HER2)-Negative Breast Cancer}} **Official Regulatory Title:** {{A Phase Ib/

II, Open-Label, Multicenter, Randomized Umbrella Study Evaluating the Efficacy and Safety of Multiple Immunotherapy-Based Treatment Combinations in Patients with Hormone Receptor-Positive Breast Cancer}}

12. Clinical Framework (Breast Cancer Specific)

Disease Area: {{Breast Neoplasms}} **Preferred UMLS Name:** {{Estrogen Receptor Positive HER2 Negative Breast Carcinoma}}
Primary Condition: {{HR-Positive, HER2-Negative Breast Cancer}}
Diagnosis Threshold: {{Disease progression during/after first- or second-line hormonal therapy with a CDK4/6 inhibitor}} **Medical Methodology:** {{Immunotherapy-based combination treatments in an adaptive umbrella framework}} **Optimization Target:** {{Objective Response Rate (ORR) and sustained Clinical Benefit Rate (CBR)}}
Ontology Codes: MeSH: {{D001943}} | HPO: {{HP:0100013}} | SNOMED ID: {{126926005}}

13. Study Procedures & Methodology

Management Pathway: {{Combination experimental therapies vs. Standard-of-care monotherapy (Fulvestrant)}} **Education Protocol (HCP):** {{Identification and management of immune-mediated adverse events}} **Education Protocol (Patient):** {{Trial stage switching (Stage 1 to Stage 2) counseling and AE reporting guidelines}}
Quality Audit Mechanism: {{Routine safety run-ins for initial cohorts (e.g., first 6 participants in a triplet arm) prior to full expansion}}

14. Observation & Follow-Up Schedule

Total Duration: {{Until loss of clinical benefit or unacceptable toxicity}} **Onsite Visit Cadence:** {{Typically Days 1 and 15 of every 28-day cycle, or Day 1 of every 21-day cycle}} **Remote Monitoring Frequency:** {{Protocol-defined intervals for safety and survival follow-up}} **Checklist Review Interval:** {{Tumor response assessments every 8-12 weeks per RECIST v1.1}}

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				